

Parametric, Polar, and Vector-Valued Calculus

Answer Key

AP Calculus BC

Quick Answers

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|------------------------------------|------------------------|
| 1. $\frac{9}{4}$ | 6. 3.534 |
| 2. 9.849 | 7. $(-t^2 + 2t)e^{-t}$ |
| 3. 6.787 | 8. 14.077 |
| 4. $\frac{\pi}{3}, \frac{5\pi}{3}$ | 9. 1.440 |
| 5. 2 | 10. — |
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1. $\frac{dy}{dx}$ at $t = 2$ for $x = t^2 + 1$, $y = t^3 - 3t$. The parametric slope is the ratio of the two t -derivatives, not a derivative of y alone.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2 - 3}{2t}, \quad t = 2 : \quad \frac{12 - 3}{4}$$

$$\boxed{\left. \frac{dy}{dx} \right|_{t=2} = \frac{9}{4}}$$

2. Speed at $t = 2$ for $\mathbf{r}(t) = \langle t^3 - 3t, t^2 \rangle$. Speed is the magnitude of the velocity vector.

$$\mathbf{v}(t) = \langle 3t^2 - 3, 2t \rangle, \quad \mathbf{v}(2) = \langle 9, 4 \rangle$$

$$|\mathbf{v}(2)| = \sqrt{9^2 + 4^2} = \sqrt{97} = 9.8488 \dots$$

$$\boxed{\sqrt{97} \approx 9.849}$$

Watch the vocabulary: velocity is the vector, speed is its magnitude, and $\frac{dy}{dx} = \frac{4}{9}$ at this instant is a third, different quantity.

3. Total distance for $\mathbf{r}(t) = \langle t^2, \frac{2}{3}t^3 \rangle$ on $[0, 2]$. Distance travelled is the integral of speed — always the integral of $|\mathbf{v}|$, never of a component.

$$\mathbf{v}(t) = \langle 2t, 2t^2 \rangle, \quad |\mathbf{v}(t)| = \sqrt{4t^2 + 4t^4} = 2t\sqrt{1 + t^2} \quad (t \geq 0)$$

$$\int_0^2 2t\sqrt{1 + t^2} dt = \left[\frac{2}{3} (1 + t^2)^{3/2} \right]_0^2 = \frac{2}{3} (5\sqrt{5} - 1) = 6.7869 \dots$$

$$\boxed{\frac{2}{3}(5\sqrt{5} - 1) \approx 6.787}$$

The substitution $u = 1 + t^2$, $du = 2t dt$ makes the antiderivative immediate — worth spotting, since the same integrand returns in the study guide's try-it.

4. Intersections of $r = 2 + 2 \cos \theta$ and $r = 3$. Set the two radius expressions equal, since a shared point needs the same r at the same θ .

$$2 + 2 \cos \theta = 3 \implies \cos \theta = \frac{1}{2} \implies \theta = \frac{\pi}{3}, \frac{5\pi}{3} \text{ on } [0, 2\pi)$$

$$\theta = \frac{\pi}{3}, \theta = \frac{5\pi}{3}$$

Caution worth teaching: solving $r_1 = r_2$ finds intersections that occur at the *same* θ . Two polar curves can also cross at a point reached at different angles (the pole is the usual culprit), so a sketch is part of a complete answer. Here $r = 0$ on the cardioid at $\theta = \pi$, where the circle has $r = 3$, so the pole is not a shared point.

5. Horizontal tangent for $x = t^3 - 3t$, $y = t^2 - 4t$. A tangent is horizontal where $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$; if both vanish the point needs separate analysis.

$$\frac{dy}{dt} = 2t - 4 = 0 \implies t = 2, \quad \left. \frac{dx}{dt} \right|_{t=2} = 3(4) - 3 = 9 \neq 0$$

$$t = 2$$

The point itself is $(x, y) = (2, -4)$, and $\frac{dy}{dx} = \frac{0}{9} = 0$ there, confirming the horizontal tangent.

6. One petal of $r = 3 \sin(2\theta)$. Polar area is $\frac{1}{2} \int r^2 d\theta$, taken over the interval that traces the region exactly once — here 0 to $\frac{\pi}{2}$.

$$A = \frac{1}{2} \int_0^{\pi/2} 9 \sin^2(2\theta) d\theta = \frac{9}{2} \int_0^{\pi/2} \frac{1 - \cos(4\theta)}{2} d\theta = \frac{9}{4} \left[\theta - \frac{\sin(4\theta)}{4} \right]_0^{\pi/2}$$

$$\frac{9\pi}{8} \approx 3.534$$

Grading note: integrating from 0 to 2π traces all four petals and gives four times too much. The half-angle identity is what makes $\sin^2$ integrable by hand.

7. $v(t)$ for $x(t) = t^2 e^{-t}$. Product rule, then factor — the factored form is what makes the sign analysis possible later.

$$v(t) = 2t e^{-t} + t^2 (-e^{-t}) = (2t - t^2) e^{-t}$$

$$v(t) = (2t - t^2) e^{-t}$$

Since $e^{-t} > 0$ always, the sign of v is the sign of $t(2-t)$: the particle moves right on $0 < t < 2$ and left after $t = 2$.

8. Inside $r = 3$, outside $r = 2 + 2\cos\theta$. Between $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$ the circle is the outer curve, so subtract the squares in that order:

$$A = \frac{1}{2} \int_{\pi/3}^{5\pi/3} [3^2 - (2 + 2\cos\theta)^2] d\theta = \frac{1}{2} \int_{\pi/3}^{5\pi/3} (5 - 8\cos\theta - 4\cos^2\theta) d\theta$$

Using $4\cos^2\theta = 2 + 2\cos(2\theta)$, an antiderivative is $3\theta - 8\sin\theta - \sin(2\theta)$, and evaluating at the limits gives $2\pi + \frac{9\sqrt{3}}{2} = 14.0774\dots$ $2\pi + \frac{9\sqrt{3}}{2} \approx 14.077$

Grading note: the limits must be the interval on which the circle is outside the cardioid. Running from $-\frac{\pi}{3}$ to $\frac{\pi}{3}$ instead computes the region where the cardioid is outside, and returns a negative integrand.

9. Arc length of $x = t^2$, $y = t^3$ on $[0, 1]$. Arc length of a parametric curve is the same integral as distance travelled — the speed integral.

$$L = \int_0^1 \sqrt{(2t)^2 + (3t^2)^2} dt = \int_0^1 t\sqrt{4 + 9t^2} dt = \left[\frac{(4 + 9t^2)^{3/2}}{27} \right]_0^1 = \frac{13\sqrt{13} - 8}{27}$$

$\frac{13\sqrt{13}-8}{27} \approx 1.440$

Pulling t out of $\sqrt{4t^2 + 9t^4}$ is legal here because $t \geq 0$ on the interval; on an interval containing negatives it would need $|t|$.

10. Manual review — expected reasoning $\mathbf{r}(t) = \langle t^3 - 3t, t^2 \rangle$, $0 \leq t \leq 3$.

- Distance vs. displacement.** Displacement records only where the particle ends up relative to where it started; total distance adds up every bit of path, including motion that doubles back. Here $x'(t) = 3t^2 - 3$ is negative on $0 < t < 1$, so the particle does reverse horizontally, and the two quantities must differ. Correct set-ups:

$$\text{distance} = \int_0^3 \sqrt{(3t^2 - 3)^2 + (2t)^2} dt, \quad \text{displacement} = |\mathbf{r}(3) - \mathbf{r}(0)| = \sqrt{18^2 + 9^2}$$

Accept the displacement written as a vector $\langle 18, 9 \rangle$ or as its magnitude. Do *not* accept $\int_0^3 \mathbf{v}(t) dt$ as the distance.

- Speeding up at $t = 1$?** Compare the directions of velocity and acceleration: $\mathbf{v}(t) = \langle 3t^2 - 3, 2t \rangle$ and $\mathbf{a}(t) = \langle 6t, 2 \rangle$, so $\mathbf{v}(1) = \langle 0, 2 \rangle$ and $\mathbf{a}(1) = \langle 6, 2 \rangle$. Then $\mathbf{v}(1) \cdot \mathbf{a}(1) = 0(6) + 2(2) = 4 > 0$, so the speed is increasing: the particle is **speeding up** at $t = 1$. Equivalent accepted argument: $|\mathbf{v}(t)| = \sqrt{(3t^2 - 3)^2 + 4t^2}$ has a positive derivative at $t = 1$. Do not accept “acceleration is positive, so it speeds up” with no reference to direction — that is the standard error this problem targets.